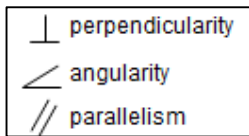


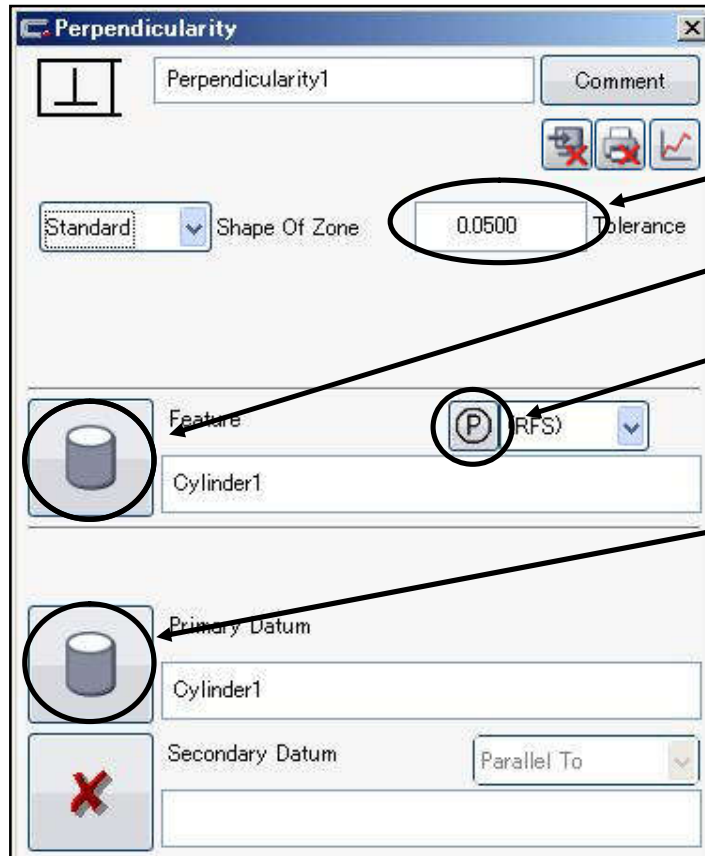
② Posture deviation

Major geometric tolerance examples



Let us evaluate whether the posture (tilt) of an object is same as the image.

You can output the result of those geometric tolerances by just adding from the menu and allocating the feature to evaluate and another feature to become the reference.



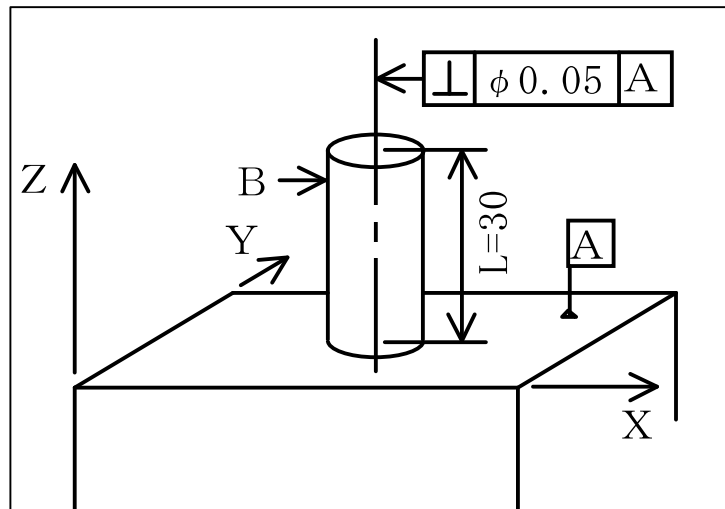
Type in the tolerance

Allocate the feature to evaluate

Setting of (P)

Allocate the datum

* To do evaluation same as the drawing, it might be required to set the evaluation length. (P)
Please refer to the Appendix 10 as for the setting method.

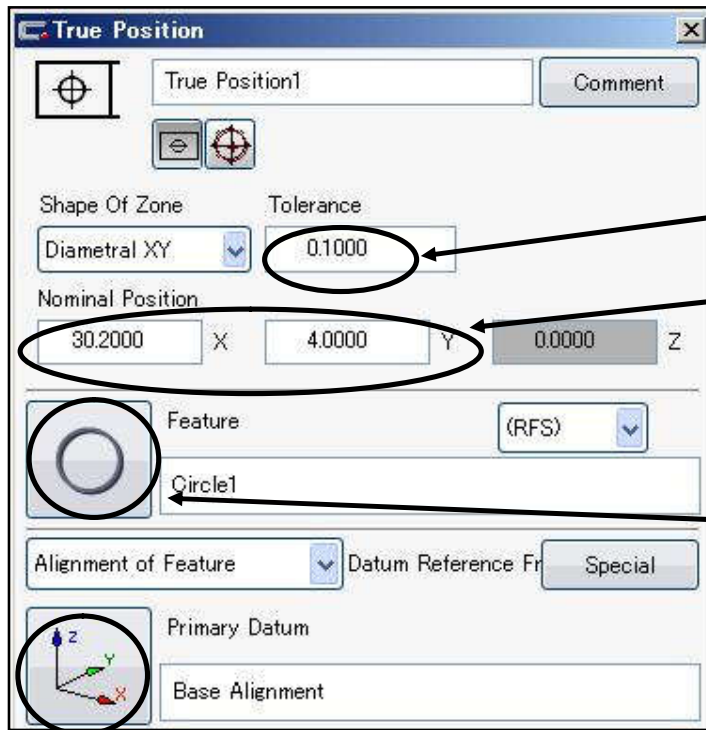


③ Position deviation

Major geometric tolerance examples

-  Concentricity
-  Coaxiality
-  Symmetry
-  True Position

Let us evaluate whether the position of an object is same as the image.
You evaluate those GD&T by adding from the menu and allocating the element to evaluate and another element to become the reference of the position.



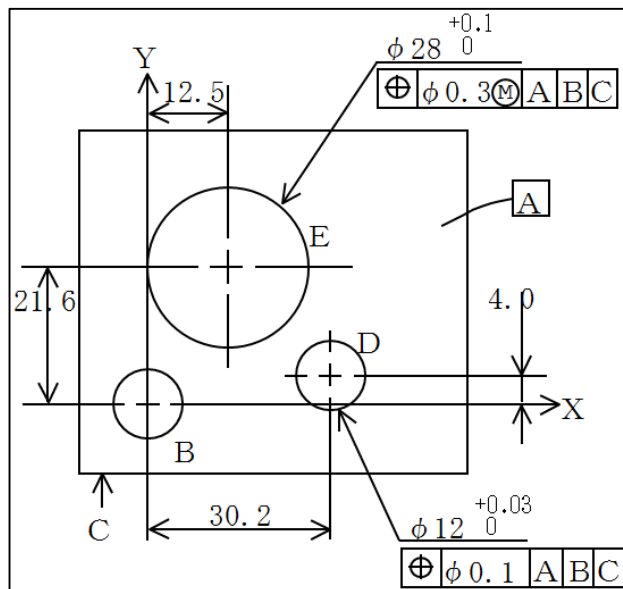
Type in the tolerance

Type in the value in the drawing in the [Nominal position] column

Allocate the feature to evaluate

In the  and , you can allocate the Base Alignment as the datum.

(Evaluation Alignment also applicable)

* Please refer to the Appendix 9 as for the setting method of .

2. GD&T Allocation Examples

① Profile deviation examples

Straightness

The screenshot shows the 'Straightness' feature control frame. It includes a straightness symbol (a line with a perpendicular tick), a text field 'Straightness1', a 'Comment' button, and three icons (a red 'X', a printer, and a graph). Below this, there is a 'Normal' dropdown menu, a 'Tol. direction' field, and a 'Tolerance' field containing '0.0500'. At the bottom, there is a 'Feature' section with a line icon and a text field '2d Line1'.

Roundness

The screenshot shows the 'Roundness' feature control frame. It includes a roundness symbol (a circle with a perpendicular tick), a text field 'Roundness1', a 'Comment' button, and three icons (a red 'X', a printer, and a graph). Below this, there is a 'Tolerance' field containing '0.0300'. At the bottom, there is a 'Feature' section with a circle icon and a text field 'Circle1'.

Flatness

The screenshot shows the 'Flatness' feature control frame. It includes a flatness symbol (a parallelogram with a perpendicular tick), a text field 'Flatness1', a 'Comment' button, and three icons (a red 'X', a printer, and a graph). Below this, there is a 'Tolerance' field containing '0.1000'. At the bottom, there is a 'Feature' section with a parallelogram icon and a text field 'Plane1'.

Cylindricity

The screenshot shows the 'Cylindricity' feature control frame. It includes a cylindricity symbol (a circle with a diagonal tick), a text field 'Cylindricity1', a 'Comment' button, and three icons (a red 'X', a printer, and a graph). Below this, there is a 'Tolerance' field containing '0.1000'. At the bottom, there is a 'Feature' section with a cylinder icon and a text field 'Cylinder1'.

② Posture deviation examples

Perpendicularity (Cylinder against plane)

The screenshot shows the 'Perpendicularity' feature control frame for a cylinder against a plane. It includes a perpendicularity symbol (two perpendicular lines), a text field 'Perpendicularity1', a 'Comment' button, and three icons (a red 'X', a printer, and a graph). Below this, there is a 'Standard' dropdown menu, a 'Shape Of Zone' field, and a 'Tolerance' field containing '0.0500'. At the bottom, there is a 'Feature' section with a cylinder icon, a 'Primary Datum' section with a plane icon and a text field 'Plane1', and a 'Secondary Datum' section with a red 'X' icon and a text field 'Parallel To'.

Perpendicularity (Plane against plane)

The screenshot shows the 'Perpendicularity' feature control frame for a plane against a plane. It includes a perpendicularity symbol (two perpendicular lines), a text field 'Perpendicularity2', a 'Comment' button, and three icons (a red 'X', a printer, and a graph). Below this, there is a 'Standard' dropdown menu, a 'Shape Of Zone' field, and a 'Tolerance' field containing '0.0500'. At the bottom, there is a 'Feature' section with a parallelogram icon, a 'Primary Datum' section with a plane icon and a text field 'Plane1', and a 'Secondary Datum' section with a red 'X' icon and a text field 'Parallel To'.

Perpendicularity (Plane against cylinder)

Perpendicularity

Feature: Plane1

Primary Datum: Cylinder1

Secondary Datum: Parallel To

Tolerance: 0.0500

Perpendicularity (Cylinder against cylinder)

Perpendicularity

Feature: Cylinder2

Primary Datum: Cylinder1

Secondary Datum: Parallel To

Tolerance: 0.0500

Angularity (Plane against plane)

Angularity

Feature: Plane3

Primary Datum: Plane1

Tolerance: 0.1000

Nominal angle: 40.0000

Angularity (Cylinder against plane)

Angularity

Feature: Cylinder1

Primary Datum: Plane1

Tolerance: 0.1000

Nominal angle: 15.0000

③ Position deviation examples

Positional tolerance

(against the Evaluation Alignment)

True Position

Feature: Circle1

Primary Datum: Alignment1

Tolerance: 0.0500

Nominal Position: X: -40.2000, Y: 32.0000, Z: 0.0000

Degree of symmetry

(against the Base Alignment)

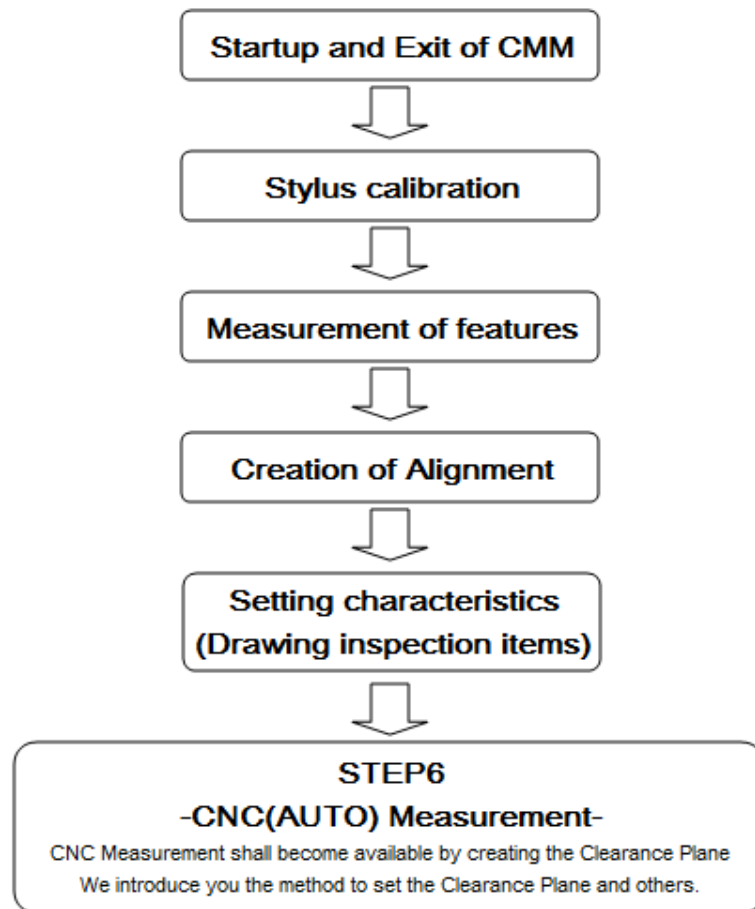
GDT Symmetry

Feature: Symmetry1

Primary Datum: Base Alignment

Tolerance: 0.0000

CNC Measurement



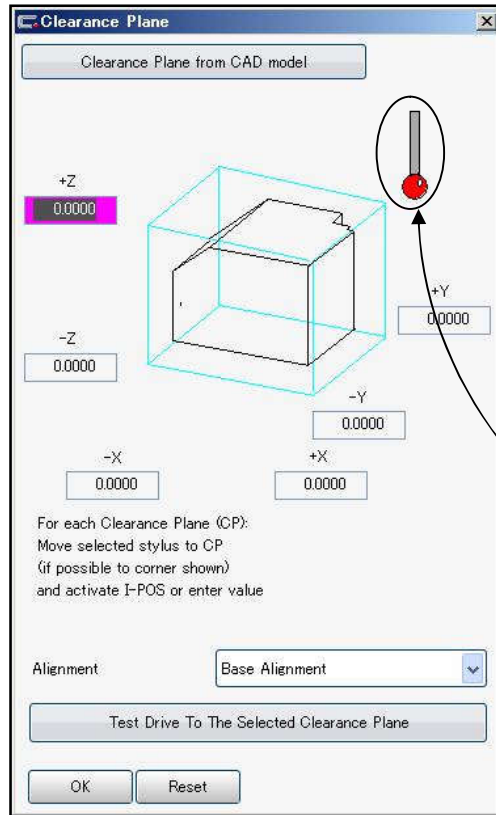
9-1 Clearance Plane Setup

It is necessary to specify a clearance plane in order to perform measurement plans (automatic measurement).

(1) The following window appears by clicking "Clearance Plane"



icon of the Prerequisites List.



(2) Using the joystick, move the stylus to the following position of the test work: about 40 mm from the upper surface; about 30 mm from the right side surface; and about 30 mm to the back (**The upper right back position to the base alignment of the work**).

(3) **Press the button on the right- joystick tip for three times** (one time for each Axis) to capture the stylus present position.

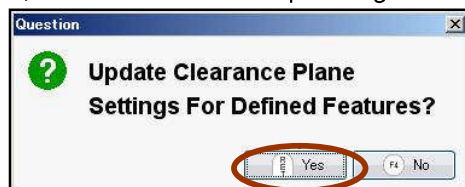
As a result, the **plus area** against the work is specified.

(4) Next, move the stylus to the **front lower left to the base alignment** of the work, place each axis about 30 mm distance from the work, and press the tip button on the right joystick for three times to capture the **minus area** for the work.

(5) Based on the above description, a cube as if it surrounds the work is displayed on the CAD screen. To display the entire view, click  (Magnification auto adjustment) of the CAD button.

(6) The following message is displayed by clicking "OK." Then, click "**Yes**."

If you click "No", it will be the same as pressing "Cancel" in the step (7).



(7) Furthermore, the following setup confirmation screen appears.

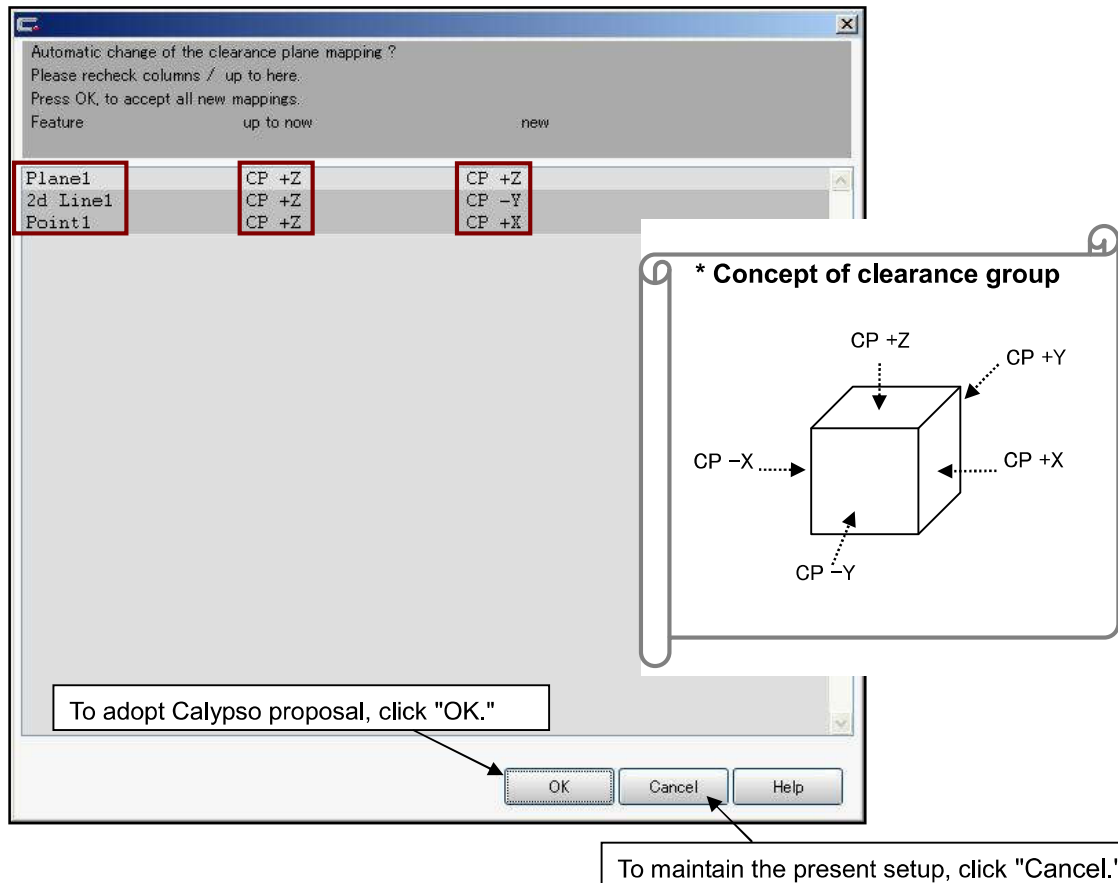
The content of this screen is as follows.

Left column: Measured feature names

Middle column: Present setup

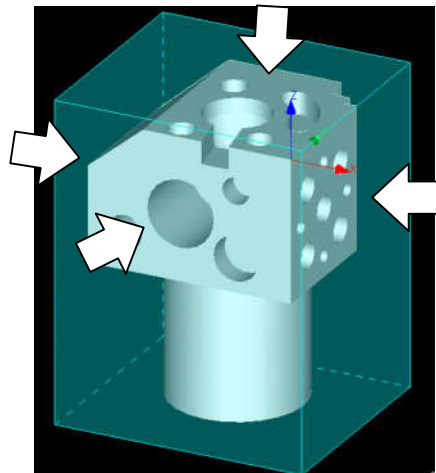
Right column: New setup proposed by Calypso as a result of performing clearance plane setup.

In addition, a clearance plane can be specified anytime by user if "Measurement Plan Editor Features" is used.



Reference!

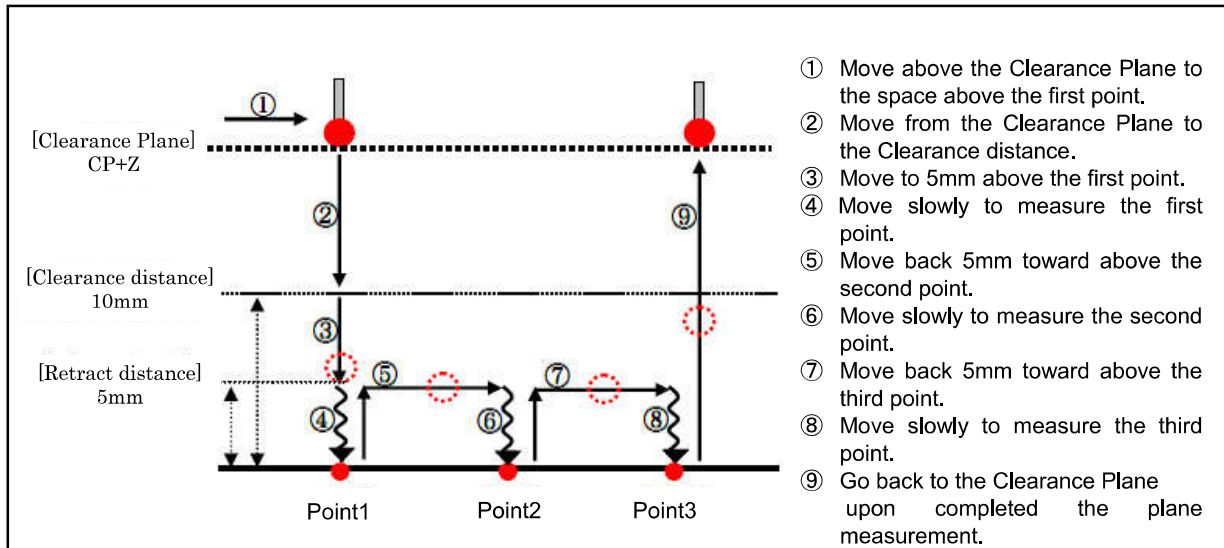
The clearance plane is specified not for the machine coordinate system but **for the alignment of a work**. Further, it can be specified by **directly inputting a coordinate value**.



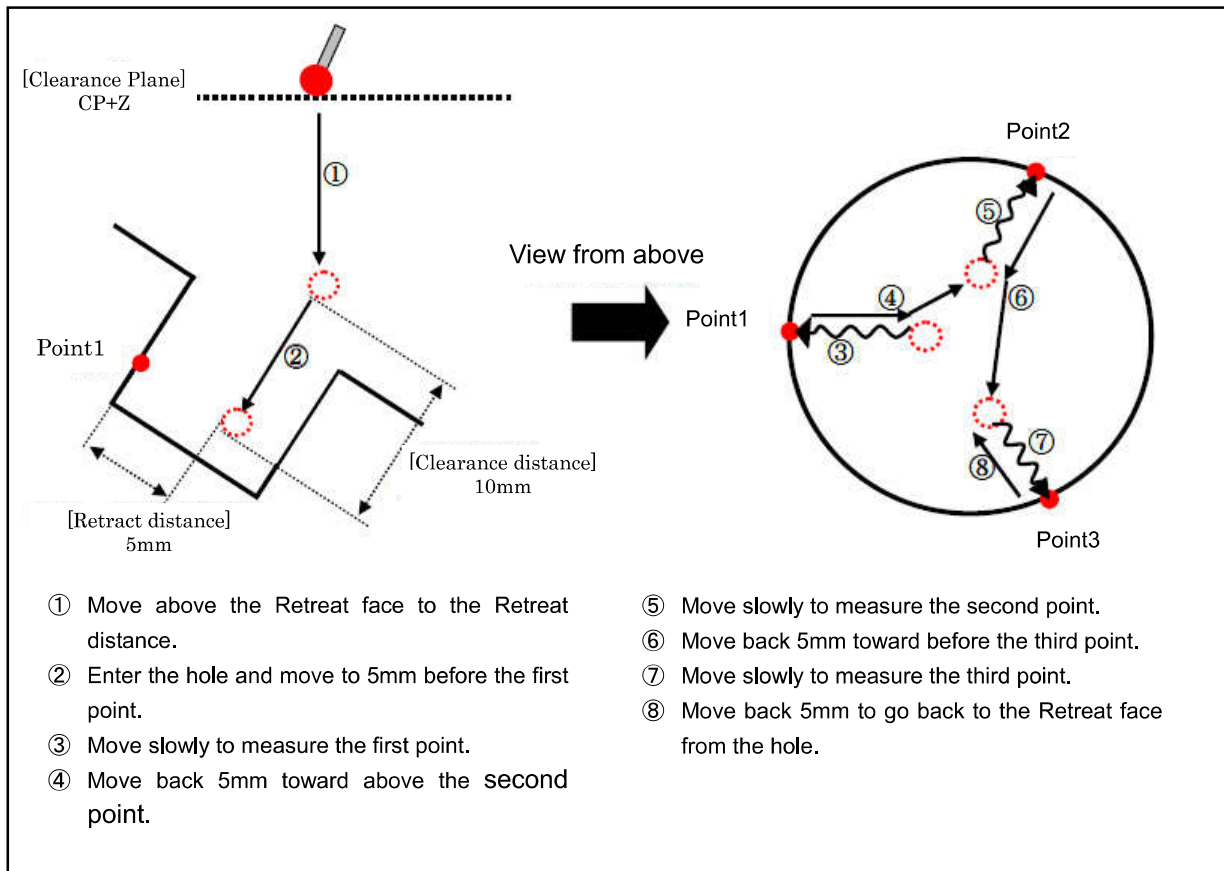
Reference: About the Move route during CNC

During CNC measurement, it moves between the features by passing through the clearance planes.
The clearance plane is built up outside the work so that the collisions can be avoided.

[To measure a plane]



[To measure / evaluate a tilted hole (it is required to know the measurement method for a tilted hole (P. 32))]

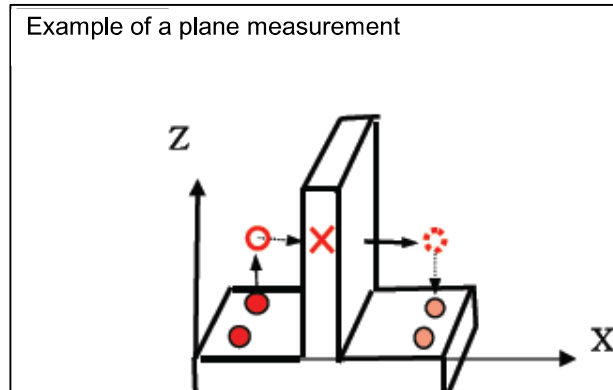


9-2 Halfway Point Setup

The movement between features was safely performed by using clearance planes.

However, the clearance plane does not affect the movement within a feature.

Therefore, it is difficult to avoid the following case.

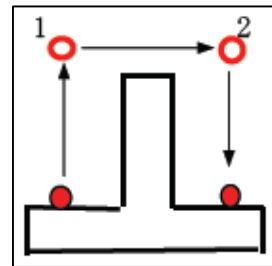


In such a case, the following measure can be considered.

In measuring a plane, after measuring the one side, a **halfway point** is specified between 1 and 2.

After moving the probe to the position intended, press the right joystick button.

A collision can be avoided because the measurement is proceeded by tracing the halfway point during CNC measurement.



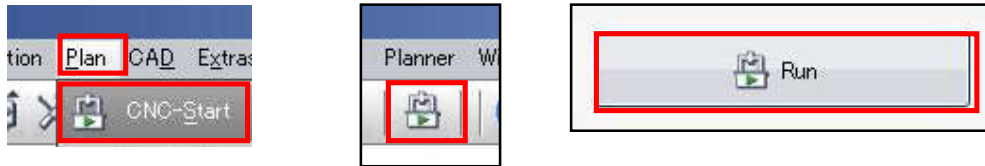
9-3 Execution of Measurement Plan

An automatic measure is performed according to the drawn-up measurement plan.

Calypso uses a clearance plane to avoid collision with a work, however, it is necessary to check whether it is safe for the measurement machine travelling around the work automatically.

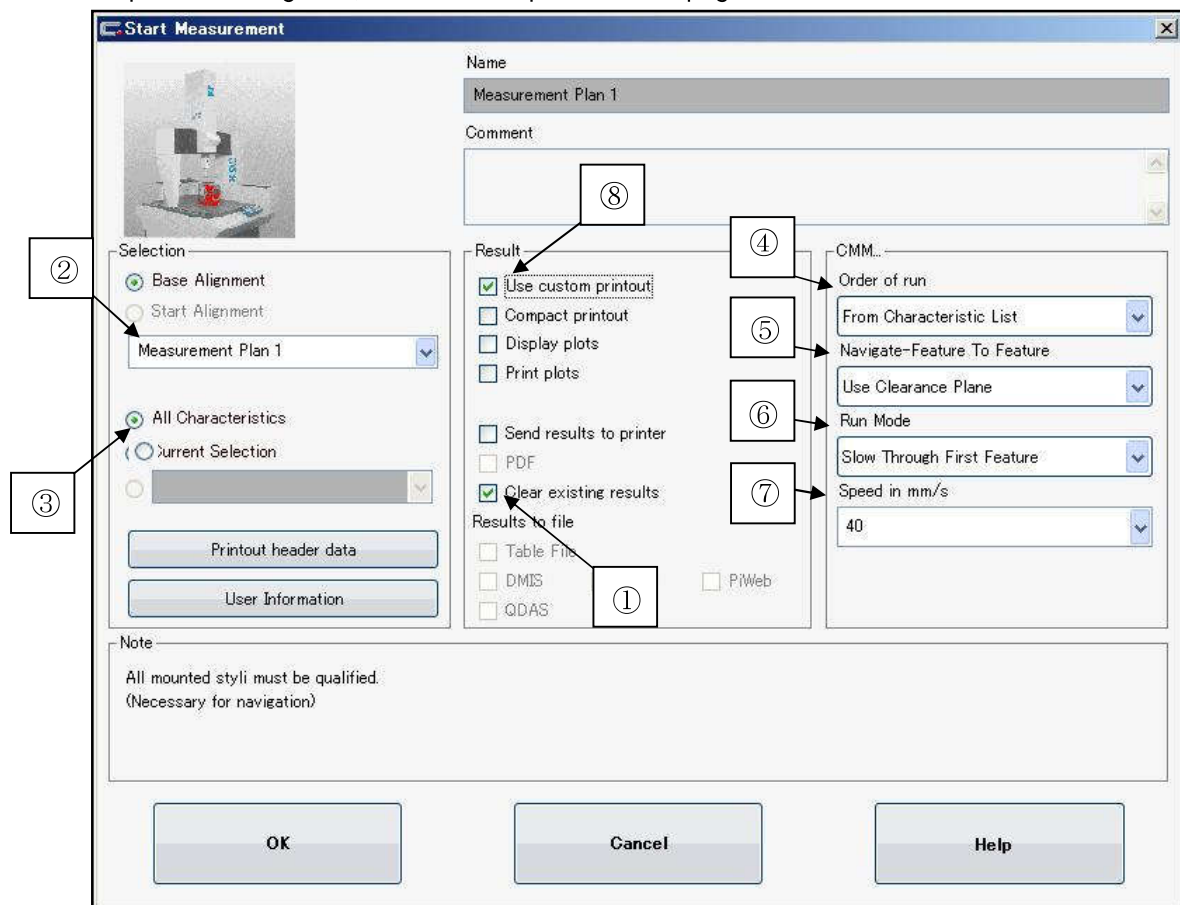
Cautions! Move the stylus to the upper part of the work.

To execute the auto measurement, you are required to select [Plan] – [Execute] from the menu bar or left click the mouse on the Exec icon in the measurement plan or CNC start button beneath the Measurement plan display area.



[Start Measurement] window appears as you have followed the above procedure.

As for respective setting item, refer to description on next page to check.



(1) Clear existing results

Calypso automatically generates the travelling route based on the reference value, and basically measures features without the actual measurement value.

For that reason, it is necessary to delete the actual measurement value previously measured in order to execute the plan.

(2) Setup of the alignment used for automatic measurement

The following three options are available here.

- Manual measurement..... It is used when the work has moved since the previous alignment setup.
It involves the step for teaching a position manually before performing the CNC measurement.
- Planned alignment..... It is used when the work has scarcely moved since the previous alignment setup.
From the base alignment, the measurement performed automatically.
This is usually used for the measurement employing jigs etc.,
- Current alignment..... It is used when the work has not moved at all since the previous alignment setup.
The base alignment is not measured.

(3) Setup of measurement target

It is selected here whether the plan is performed entirely or partially.

(4) Order of Run

The order for performing the automatic measurement is specified here.

Here, you can select the order for the feature list or the characteristics list.

(5) Run mode

Check that it is specified as "Use Clearance plane."

(* If the "Auto" is not specified, there is a possibility of collision.)

(6) Mode to perform

The way of move while performing the automatic drive is specified here.

- Slow Through first Feature : Measure the first feature in a low speed.
- Normal : Keep the speed designated at (7) above to move from the start to the end.

(7) Speed

During the automatic drive, the value specified here becomes the maximum value of the speed volume.

For performing a measurement plan for the first time, specify a low speed such as 40 (mm/sec) in order to ensure safety and avoid damages on the probe.

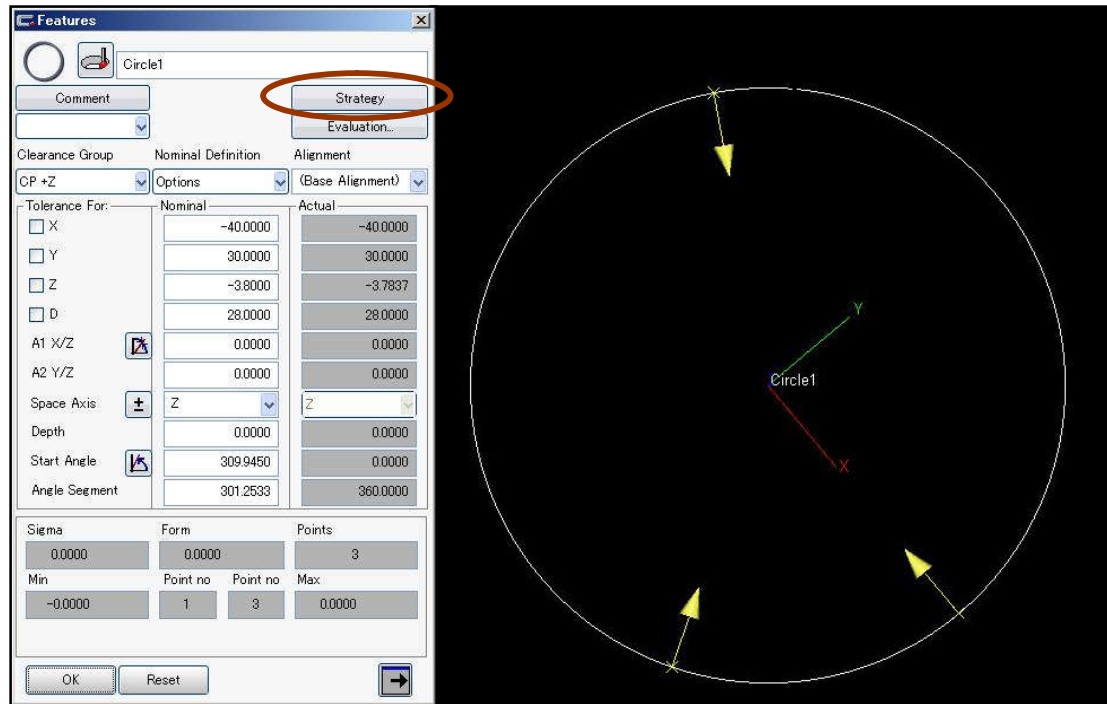
(8) Setup of output form

Finally, specify the speed volume to the minimum and press "OK."

9-4 Segment measurement of circle

It is applicable to implement multi point measurement of a circle under equal division by designated point number.

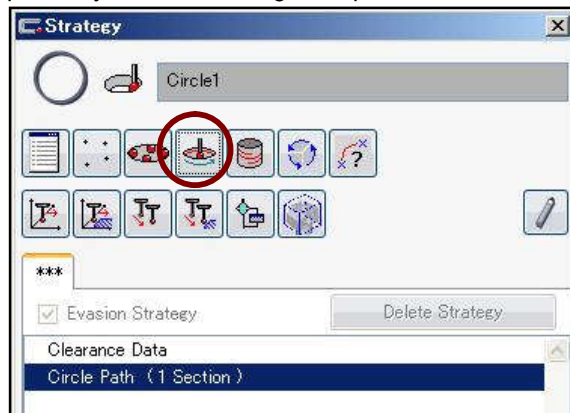
- ① Open a circle you want to do multi point measurement and left click the **Strategy**.



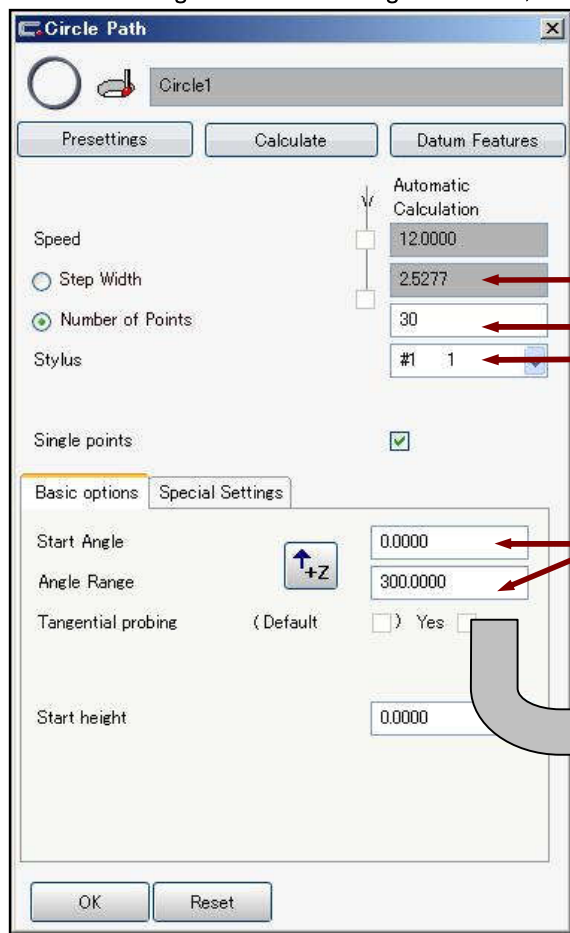
- (2) Select all the probing points and cut them.



- (3) Click the **Circle Path icon** to display "Circle Path (1 Section)",
and open it by double-clicking the option.

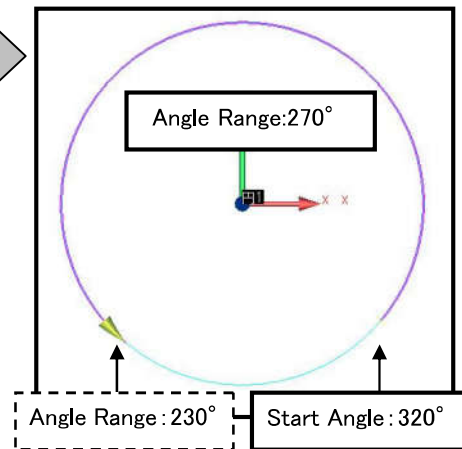


- (4) Perform each setting. When the setting is finished, click "OK" to close the circle feature.



Designate a pitch to apply or
measurement point number.
- The stylus used for Circle Path

Measurement range of the circle
Set a range of the arc if there was a
notch or the like.

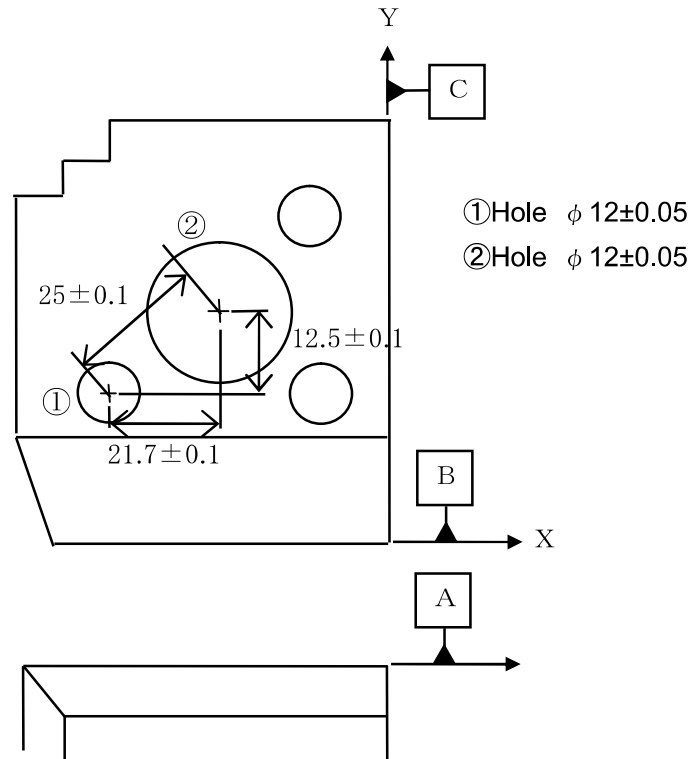


<< *Memo* >>

☆Let's measure in CNC!☆

Make a plan to automatically measure dimensions shown in the following drawing.

Note to measure a circle by the [Circle segment measurement].



(1) Click "Create New Measurement Plan"  icon, input a name, and click "OK."

(2) Perform a calibration for the stylus

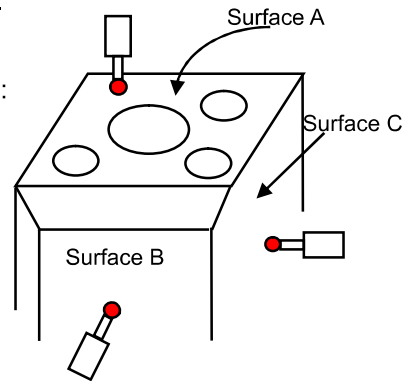
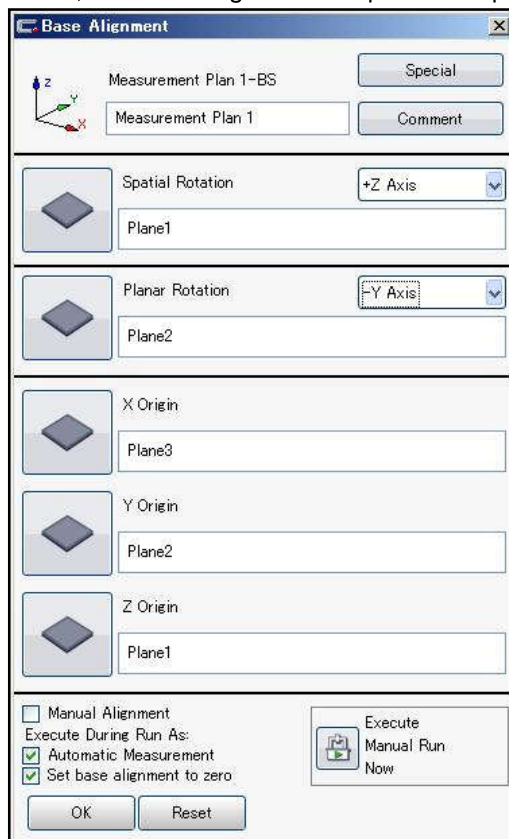
With reference to "Stylus Calibration" on Page 17 and after, add a back-facing and a left-facing styluses

(3) Base Alignment setup

Click "**Base/Start Alignment**" icon , select "Create a new base alignment", click "OK", and perform measurements in the following order.

- Measure the Surface A by the down-facing stylus, and click "OK."
- Measure the Surface B by the back-facing stylus, and click "OK."
- Measure the Surface C by the left-facing stylus, and click "OK."

As a result, the Base Alignment template is displayed as follows:



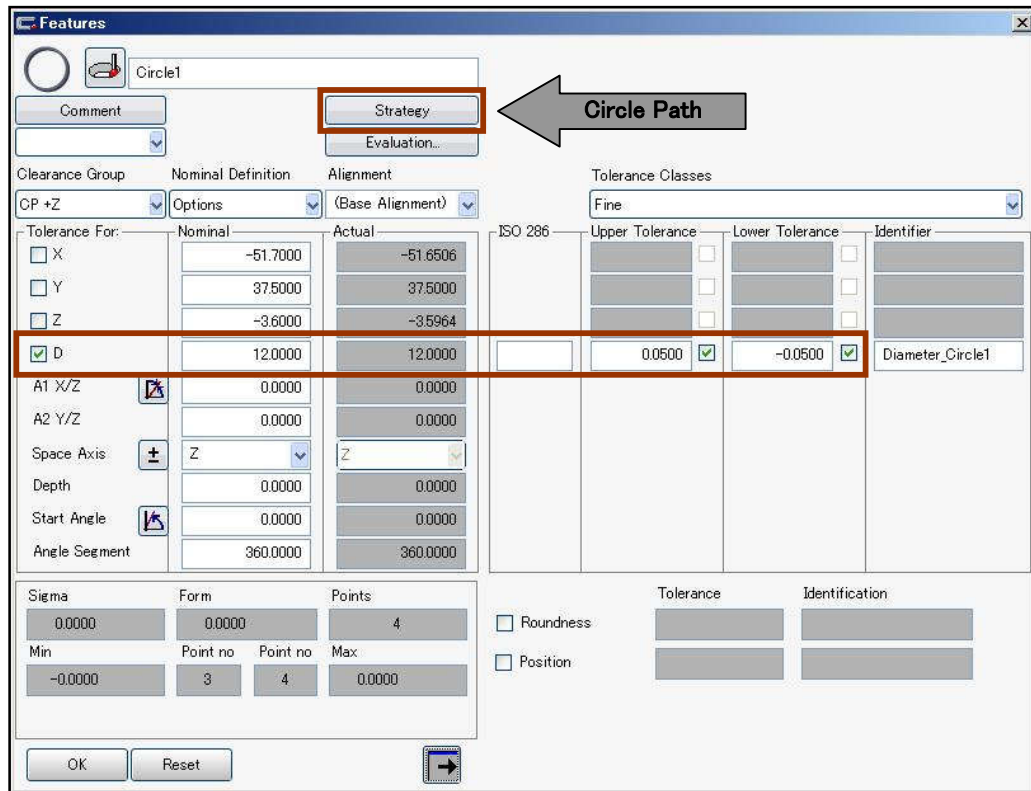
- Click "OK" to terminate the base alignment setting.

(4) Click "**Clearance Plane**" icon  . (Set the stylus back to facing down)

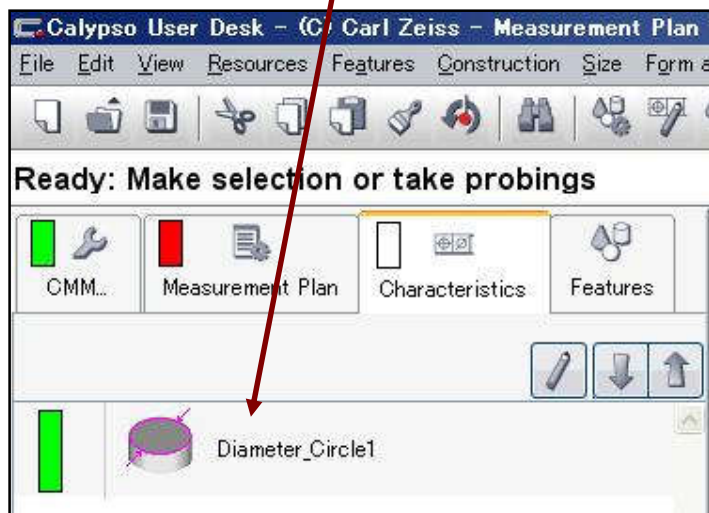
- Move the stylus to the upper right back of the test block, and press the button on the right joystick tip for three times (Plus side area).
- Move the stylus to the front lower left, and press the button on the right joystick tip for three times (Minus side area).
- A message appears by clicking "OK." Click "Yes" or "OK."

(5) Measurement

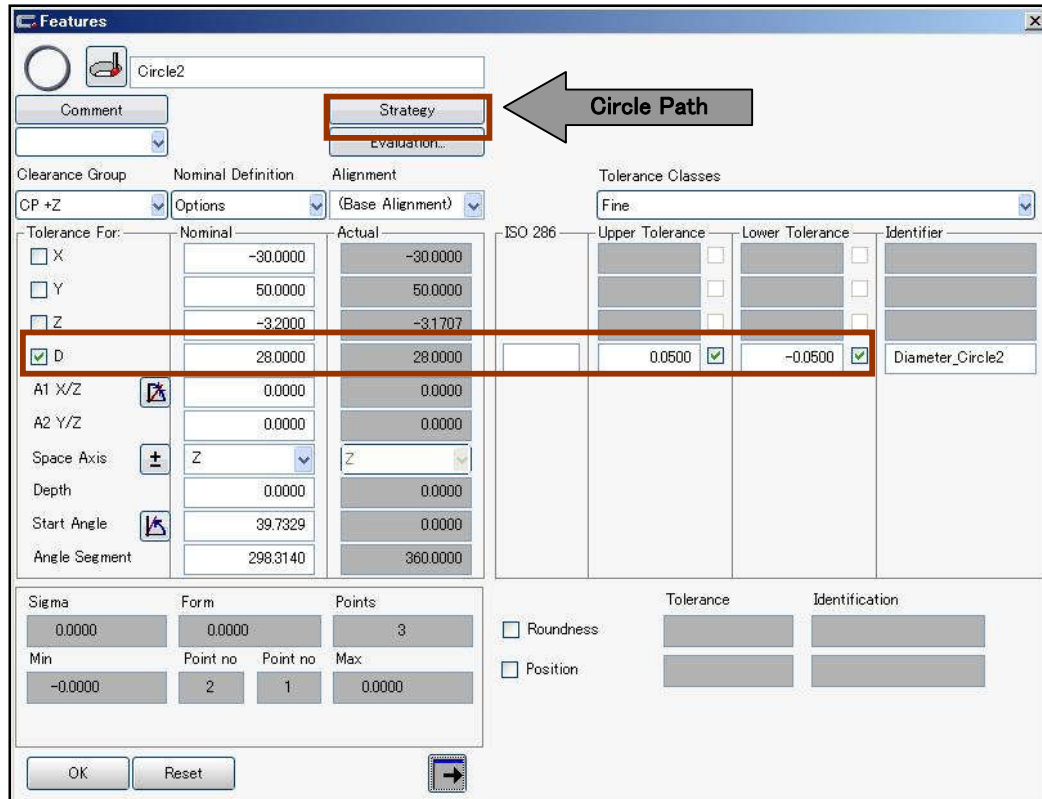
1. Perform probing for three points or more on the Hole1. Click "Diameter (D)" of The Feature Recognition template, input the nominal value and the tolerance, and click "OK" to close the window.



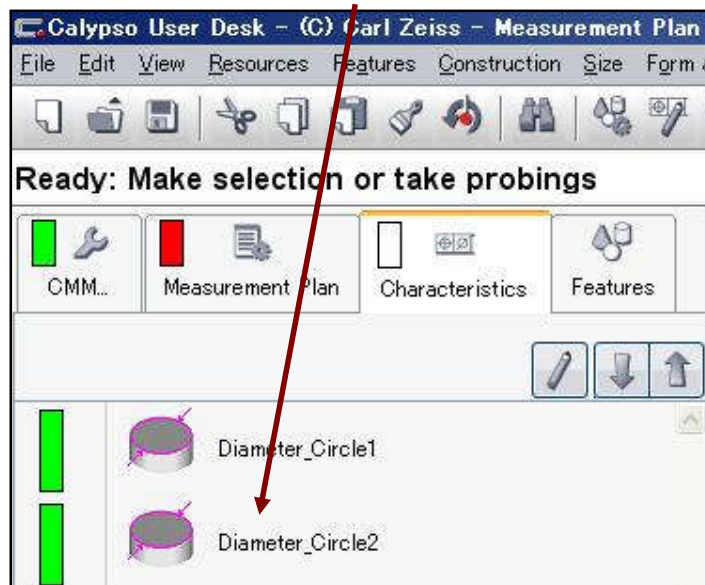
The characteristic of Diameter **Circle 1** is displayed by checking the **characteristics list**.



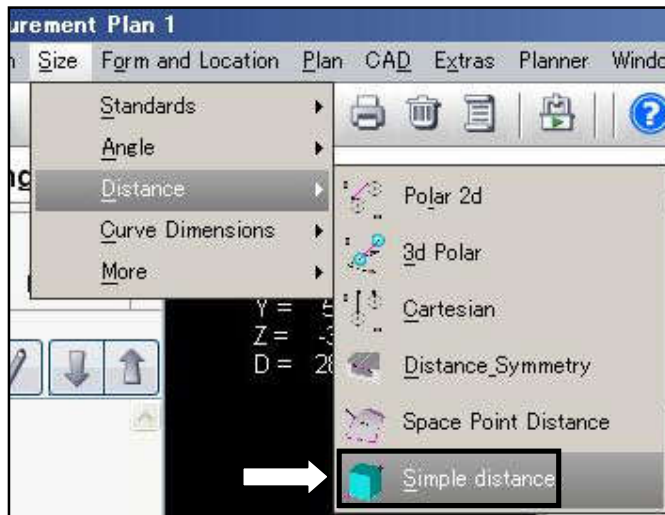
- After probing three or more points on the Hole 2, click "**Diameter (D)**" of the feature recognition template, input reference value and tolerance, and click "OK" to close the window.
Open the feature window of the circle and set the segment of the circle from the Strategy.



Diameter **Circle 2** is added by checking the **characteristics** list.

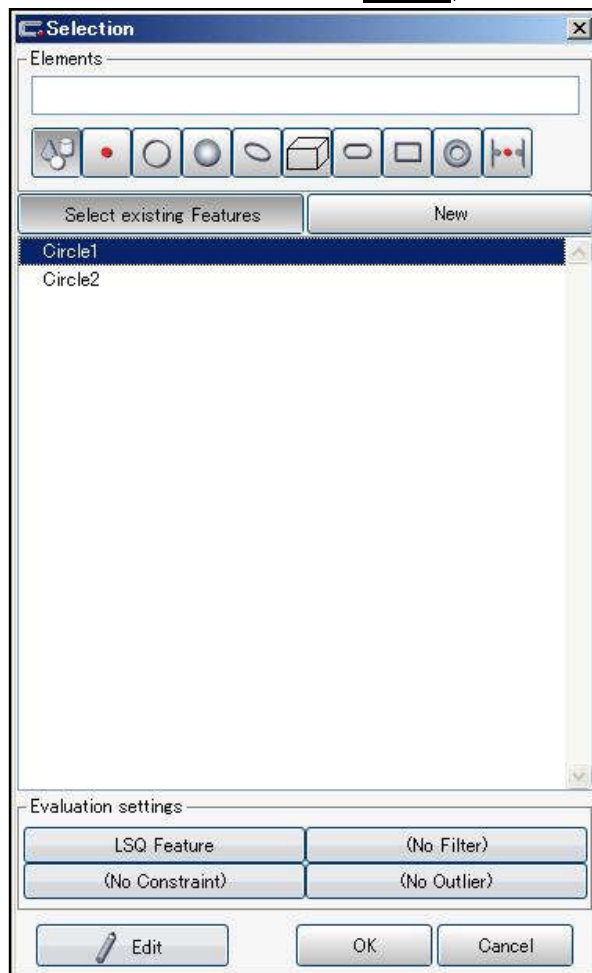


3. Next, the direct distance for (1) and (2) and coordinate distance of 21.7 plus or minus 0.1 and 12.5 plus or minus 0.1 are obtained.



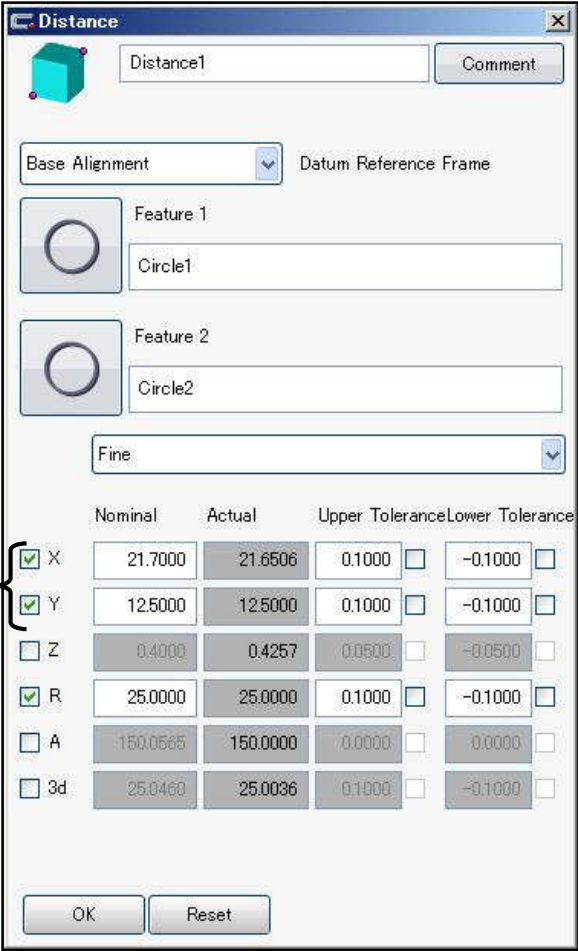
Double-click **Distance1** icon displayed on the characteristics list to define the features.

Click "Feature1" button to select **Circle1**, and click "OK."



Likewise, allocate **Circle2** to the Feature2.

The result appears below. Place checkmarks on X, Y, and R, input Nominal value and the Tolerance, and click "OK" to close the window.



The Distance dialog box is shown with the following fields and options:

- Distance1**: Text field for the distance name.
- Comment**: Text field for a comment.
- Base Alignment**: Dropdown menu.
- Datum Reference Frame**: Text field.
- Feature 1**: Circle1
- Feature 2**: Circle2
- Fine**: Dropdown menu.

The table below shows the distance results and tolerances:

	Nominal	Actual	Upper Tolerance	Lower Tolerance
☒ X	21.7000	21.6506	0.1000	-0.1000
☒ Y	12.5000	12.5000	0.1000	-0.1000
☐ Z	0.4000	0.4257	0.0500	-0.0500
☒ R	25.0000	25.0000	0.1000	-0.1000
☐ A	150.0565	150.0000	0.0000	0.0000
☐ 3d	25.0460	25.0036	0.1000	-0.1000

Annotations:

- Coordinate distance**: Points to the X, Y, and Z rows.
- Direct distance**: Points to the R, A, and 3d rows.

Buttons: OK, Reset